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Unit – 4 : Session – 1 : SLO – 1 | Trees — Introduction

■ I. Fill in the Blanks

- A TREE is a non-linear data structure used to represent hierarchical relationships.
- The topmost node in a tree is called the ROOT.
- A node that has no children is called a LEAF (terminal node).
- A node having at least one child is called an INTERNAL (non-terminal) node.
- A PARENT is a node that has one or more child nodes.
- The nodes that are directly connected and come after a node are its CHILDREN.
- The number of edges from the root to the node is called the DEPTH (level).
- The length of the path from a node to the deepest leaf is called the HEIGHT.
- A tree with no nodes is called an EMPTY (NULL) tree.
- A tree in which each node has at most two children is called a BINARY tree.
- The degree of a node is the number of its CHILDREN.
- The ANCESTORS of a node includes all the nodes along the path from the node to the root.
- A tree in which all leaf nodes are at the same level is a PERFECT BINARY TREE.
- The tree data structure is used in many applications such as FILE SYSTEMS, DATABASE INDEXING and COMPILER DESIGN.
- The total number of nodes in a full binary tree is $2^{(h+1)} - 1$, where h is the height.